Safi Ullah

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RESEARCH INTERESTS

I am broadly interested in the robustness and generalization of machine learning systems under distribution shift, with a focus on transfer learning and domain adaptation across sequential and relational data. My approach draws on attention and graph based architectures, with an emphasis on building trustworthy AI for high stakes and critical infrastructure systems in real-world settings.

EDUCATION

Bachelor of Science in Computer Science — *Nov 2021 – Nov 2025*
Quaid-i-Azam University - Islamabad, Pakistan — *CGPA: 3.60 / 4.00*

Relevant Coursework: Algorithms & Data Structures, Artificial Intelligence, Machine Learning, Operating Systems, Database Systems, Software Engineering, Computer Graphics, Web Development. Final-Year Project: StartLnk: AI-powered startup networking platform; led product development, platform architecture, and deployment.

RESEARCH EXPERIENCE

Researcher - Cross-Market Equity Transfer Learning — *Dec 2025 – Present*
Quaid-i-Azam University, Islamabad, Pakistan

- Designed a six-stage end-to-end ML pipeline for cross-market transfer learning between India (NSE/BSE, 16 companies) and Pakistan (PSX, 15 companies) across four sectors, covering 125,668 daily observations over a 23-year window (2000–2023).
- Proposed a **CNN-BiLSTM-Attention** architecture with causally-padded convolutions, correcting temporal-leakage issues present in standard zero-padding implementations used by prior financial forecasting systems.
- Identified a **first-documented directional transfer asymmetry** (+2.06 pp, Pakistan → India) through five-scenario evaluation framework: full-market pooling, sector-conditioned 4×4 Transferability Matrix, and progressive fine-tuning experiments (10–50% target-market data).
- Applied Maximum Mean Discrepancy (MMD) domain-shift analysis across all 16 sector combinations; confirmed near-random directional accuracy consistent with semi-strong Efficient Market Hypothesis in South Asian emerging markets.
- Validated findings with **Friedman test**, **Bonferroni-corrected Wilcoxon signed-rank tests** ($n = 31$), and **Cohen's d** effect sizes, ensuring statistical rigor suitable for peer-reviewed submission.

PUBLICATIONS & MANUSCRIPTS

S. Ullah. "Cross-Market Transfer Learning for Equity Price Prediction: An Empirical Study of India–Pakistan Transfer Dynamics." *Manuscript Under Preparation.*

PROFESSIONAL EXPERIENCE

Software Engineer — *Jun 2025 – Present*
National Aerospace Science & Technology Park (NASTP) - Islamabad, PK

- Engineered Python-based automation tools for aerospace and critical protection systems, streamlining multi-step operational workflows and reducing manual intervention overhead.
- Integrated on-premise deep learning and AI models into mission-critical applications, enabling intelligent decision support without external API dependencies.
- Contributed to Qt QML/C++ desktop application development for high-priority security systems, focusing on performance, security, and maintainability requirements.

AI Engineer Intern — *Jun – Jul 2024*
DTtools AI - Islamabad, PK

- Developed AI-powered applications integrating computer-vision and NLP models via REST APIs; deployed interactive user interfaces using Streamlit, Tkinter, and Flask.
- Built a computer-vision pipeline classifying user mental states from eye-feature biomarkers using TensorFlow and OpenCV, contributing to a live product feature.

Software Intern *Jul – Aug 2023*
Technevity - Islamabad, PK

- Developed an Interview Management System using ASP.NET MVC, C#, and Entity Framework; optimised SQL Server queries to improve HR data retrieval performance.

TEACHING EXPERIENCE

Teaching Assistant - Introduction to Artificial Intelligence *Oct 2024 – Feb 2025*
Quaid-i-Azam University - Islamabad, PK

- Designed and delivered lab sessions on AI search algorithms (A*, UCS, BFS/DFS), supervised/unsupervised learning (SVM, Random Forest, K-Means, PCA), and deep learning fundamentals (backpropagation, regularisation) in Python and TensorFlow.
- Mentored **30+ undergraduate students** through complete ML workflows — data preprocessing, cross-validation, hyperparameter tuning, and performance evaluation.

PROJECTS

StartLink - AI-Powered Startup Ecosystem Platform [Web →]
Python · Django REST Framework · React.js · PostgreSQL · TensorFlow Lite · Docker

- Architected a full-stack platform connecting entrepreneurs, investors, and collaborators; implemented an **AI recommendation engine** (scikit-learn, TF Lite) surfacing relevant startups based on interaction history and interest profiles.
- Built scalable REST APIs, real-time messaging, investor-matching, and team-formation workflows with Django backend and React + Tailwind CSS frontend.

Cross-Network Intrusion Detection with Graph Neural Networks [GitHub →]
Python · PyTorch Geometric · GraphSAGE · Domain Adaptation · NetFlow v2

- Modeled network traffic as flow graphs and trained an edge-feature GraphSAGE classifier achieving high within-dataset detection accuracy across NF-UNSW-NB15 and NF-CSE-CIC-IDS2018 benchmark traffic.
- Quantified severe cross-network performance degradation under naive transfer and applied adversarial domain adaptation to recover detection accuracy, validating gains with Friedman, Wilcoxon, and Cohen's d statistical tests.

Virtual Try-On Web Application [GitHub →]
Python · Hugging Face Diffusion Models · Streamlit · Transfer Learning

- Fine-tuned a diffusion-based virtual garment try-on model to improve photorealistic fitting accuracy; deployed as an interactive Streamlit application with real-time image upload and garment selection.

Eye-Gaze Mental State Predictor [GitHub →]
Python · TensorFlow · OpenCV · Computer Vision

- Trained a supervised deep-learning classifier from scratch to predict mental states (stress / calm) from eye-feature biomarkers; applied advanced preprocessing and augmentation pipelines to optimise classification accuracy.

TECHNICAL SKILLS

Programming Languages	Python, C++, C#, Java, JavaScript, TypeScript, HTML/CSS
AI / ML & Deep Learning	TensorFlow, Keras, PyTorch, scikit-learn, Hugging Face Transformers, LangChain, OpenCV
Data Science & Statistics	NumPy, Pandas, Matplotlib, Seaborn, SciPy, Hypothesis Testing, Effect-Size Analysis, MMD
Web Technologies	Django REST Framework, Flask, React.js, Tailwind CSS, ASP.NET MVC, REST APIs, Streamlit
Databases	PostgreSQL, MySQL, Firebase, SQL Server, SQLite
Tools & Platforms	Git, Docker, Jupyter, Google Colab, Linux/Bash, VS Code, Overleaf/LaTeX

HONORS & AWARDS

Certificate of Appreciation — Pakistan Japan Centre for Mutual Cooperation *Apr 2023*
Awarded for completion of Data Science Workshop: supervised/unsupervised ML algorithms and cloud-based model deployment.

CERTIFICATIONS & TRAINING

Machine Learning Specialization (Supervised, Advanced, Unsupervised Learning) — Andrew Ng, Coursera *2023*
LangChain for LLM Application Development — DeepLearning.AI *2023*
ChatGPT Prompt Engineering for Developers — DeepLearning.AI *2023*
Certified Cloud Applied Generative AI Engineer — PIAIC, Pakistan *2023–2024*